Path Planning And Trajectory for UAVs

Tarek Uddin Ahmed
170431981
Umair Bilal Chaudhry
MSc Artificial Intelligence

***Abstract*— The application of a fusion path planning algorithm based on the combination of a Genetic Algorithm (GA) with the Ant Colony Optimization (ACO) algorithm to aid UAV data collection for dynamic environments has shown to be beneficial when compared against a UAV set on a defined trajectory. Detouring is effective when the UAV has a communication range of 100m, where the UAV can directly capture 7.89% more nodes than the UAV set on a defined trajectory and 14.56% more nodes with caching enabled in a dynamic environment of 100 nodes. When comparing the fusion algorithm to the GA and ACO with TSP, the fusion algorithm produces a path 42.98% shorter than the GA and 8.76% shorter than ACO. The ACO part of the fusion algorithm is also 44.9% faster than the standard ACO algorithm suggesting GA help improve the performance of ACO.**

Keywords—UAVs; Ant Colony Optimization; Genetic Algorithm; Path Planning; Trajectory Planning

I. Introduction

UAVs play an important role in creating an energy-efficient way of collecting data for Wireless Sensor Networks (WSN). Before the application of UAVs for data collection, massive data collection from WSNs was inefficient due to the network relying on multi-hop forwarding to transmit data. In addition, due to lacking central servers, WSNs are decentralized and have unreliable wireless links, contributing to the network's unreliability for massive data collection. However, this makes WSNs ideal for deployment to disaster and conflict environments where infrastructure may be damaged. UAVs can allow for efficient data transmission due to the ability to carry gear such as sensors, cameras, transmitters, and receivers. They can enable data transmission to nodes outside of the communication range due to being able to move to and from the base station. UAVs demonstrate the potential to improve WSNs, but the problem arises when planning a path since UAVs run on limited resources with limited computational power. Research has been conducted to create lightweight algorithms with real-time operation capabilities to allow for replanning while the UAV is on the route to optimize for the shortest path. (Goudarzi *et al.*, 2019)

UAVS can also be used to collect data from wireless sensor networks in wide areas and due to its mobility, can also increase network coverage. Nodes transmit data via the multi-hop transmission relay method, which can cause increased power usage and data loss if the multi-hop transmission fails. Making all nodes transmit to the UAVs instead of transmitting to a fusion center will counter the issues related to multi-hop transmission. UAVs are adaptive and can have various use cases such as relays and base stations, and since modelling flight propulsion based on energy consumption is difficult to optimize, flight time minimization can also benefit energy consumption. Path and trajectory planning is required for UAVs as they are resource and restricted, so they must follow an optimal route to have a quicker flight time and maximum efficiency. (Gong *et al.*, 2018)

Research on the application of UAVs to monitor the transmission lines in the rapidly growing Chinese Network infrastructure, which is increasing by 5% per year as of 2020. The existing equipment of the transmission lines is susceptible to environmental-induced damage such as plant invasions and man-made and animal-induced damage. The existing monitoring methods involve manual labor, which is inefficient in carrying out the task as the cost of resources is high, and the job can be dangerous to health, where UAVs do not succumb to the same disadvantages allowing it to be the choice when inspecting transmission lines. (Yin *et al.*, 2021)

UAVs have been deployed into areas where the environment may not be known, which can lead to dangers. A path planning method is required to generate a suitable trajectory safe for both the UAV and the environment. (Wang, Jiang and Bao, 2021). Path planning is an important aspect of UAV operation where the quality of the path planning algorithm dictates the efficiency of the UAV's performance, creating a safe and collision-free path from the starting point and end goal. Traditional methods such as A*, Dijkstra and D* can find the optimum solution, such as the shortest path but can be computationally expensive. (Chen and Qi, 2020)

The Travelling Salesman Problem (TSP) can be used to test the efficiency of Path Planning algorithms due to the problem being a Non-deterministic Polynomial (NP). Path Planning methods must be optimal due to UAVs' restrictions, such as limited load and battery. Wireless charging towers will allow UAVs to increase their range while reducing pollution and helping the environment. Using the brute-force method to solve TSP can increase the algorithm's time complexity exponentially when increasing the number of nodes, motivating researchers to invent methods to obtain optimal solutions while decreasing the time complexity. (Xu and Che, 2019)

UAVs have the potential to improve efficiency in the industry though the real world is unpredictable. There may be obstacles, weather issues, and hardware issues with UAVs, such as engine failures or human errors such as not refueling the UAV before take-off resulting in a defined trajectory being inefficient. This paper demonstrates the application of a path planning algorithm to aid the UAV in its node capturing task in a dynamic unknown environment of nodes that move using levy motion. This paper also aims to make UAV data collection more efficient by allowing the UAV to be more autonomous on the path it takes to collect data.

II. Related work

Conventional Path Planning methods are computationally expensive. Soft computing techniques such as Genetic Algorithms are stochastic, motivating researchers to apply them in solving the path planning problem. Genetic Algorithms do not require any derivative information or initial estimates as the methods can search the entire solution space resulting in a higher probability of finding the global optimum.(Nagib and Gharieb, 2004).

Genetic Algorithms overcame issues such as falling into local minimum as seen by the potential field method and also times and space complexity issues displayed by the A* algorithm. Genetic Algorithms are designed to carry out a global search by optimizing globally and can be characteristics of implicit parallelism, which has shown to be effective at solving complex problems that traditional methods have failed to solve. (Cao *et al.*, 2016). Meta-heuristic methods such as Genetic Algorithms and Ant Colony Optimization algorithms search globally, allowing them to come out of local optimums while being computationally efficient since they approximate near-optimal solutions. (Chen and Qi, 2020)

Genetic Algorithms are a global search method inspired by the theory of evolution (Natural Selection). The algorithm starts with a randomly generated initial Population which consists of Chromosomes and Genes. Chromosomes are solutions ranked by their fitness value, which displays how good the candidate solution is. The Key idea of Genetic Algorithms is the ability to explore the search space in parallel without the need to have a differential or optimized function where the solution's quality depends on the Chromosomes' length. Genetic algorithms tackle the path planning problem by optimizing for the shortest path by being given a starting position, goal position and a 2D map of the environment to plan a collision-free path between them where chromosomes are possible routes and genes are the nodes. (Nagib and Gharieb, 2004)

Liu et al. used a Genetic Algorithm with the K-means clustering algorithm to create a method called MTSP, and Wang et al. improved the Genetic Algorithm by implementing double chromosome coding called MDTSP, both of which tackled the multi travelling salesman problem but are incapable of multi-UAV synchronised access to multiple goals at the same time.(Yin *et al.*, 2021)

(Yin *et al.*, 2021) propose a novel method called CH&DS-GA to solve the issues faced by MTSP and MDTSP by implementing a TSP-Drones model with charging stations to mitigate power-related issues. The key idea is an enhanced Genetic Algorithm with the implementation of a dual population selection system and the use of a multi-level hybrid coding for the chromosomes. Usually, the chromosomes are the individual routes in a path planning problem.

(Yin *et al.*, 2021) also implements multi-level encoding containing the UAV number, charging station number with 0 meaning no charge required, and task number indicating the next node to visit. CH&DS-GA is tested against an exhaustive system with less than ten nodes to demonstrate its performance and effectiveness.

(Goudarzi *et al.*, 2019) propose a novel Path Planning method called BL-TSP to solve UAV-aided data collection issues. The key idea is to generate the UAV trajectory from TSP followed by the application of the path smoothing algorithm, compensating the UAVs maneuverability limitations such as turning capabilities where the algorithm uses circles with a tunable parameter R to turn UAVs preventing sharp turning and smooth paths.

(Xiang Rong et al. 2020) test the path planning algorithm on a two-dimensional map of the Raytheon hospital created using the raster method, using white squares to signify free space and black squares to represent obstacles. The key idea is to use a Genetic Algorithm to find the coordinate point that is globally optimal from a set of local coordinates and set as the target for the A* algorithm, which finds the shortest path to the current UAV position resulting in a faster algorithm that requires less memory. (Xiang Rong and Hong Mei, 2020)

(Xiang Rong et al. 2020) use 32 coordinates for the first population of the Genetic Algorithm to keep the coordinates near the starting point, improving the algorithm's computational speed. The proposed algorithm uses the roulette wheel selection method to select the population where a path with a higher fitness has a higher probability of being chosen. Crossover happens if the randomly generated number is less than the judgement value (pm), where a higher value will result in no crossover or mutation. The algorithm is more efficient than the base genetic algorithm and saves storage, improving the efficiency of path planning though there is still a problem of path smoothness that needs further research.(Xiang Rong and Hong Mei, 2020)

The moderation function calculates distances between the individual coordinates in the population and the distance between the start and end goal. The function also removes the worst coordinates points when the crossover function is odd, allowing the pool to become even since the crossover function works when the pool has an even number of coordinates. (Xiang Rong and Hong Mei, 2020)

The A* algorithm searches eight nodes starting from the start node while aiming for the target node, and for every 0.5 distance searched, the coordinate is compared to the constraints of the moderate function. The algorithm compares the current coordinates and the global optimum coordinates when the coordinates satisfy the moderate function, and if the current coordinates are optimal, the search is stopped, replacing the target coordinates generated from the Genetic algorithm. If the conditions have not been met, the search continues, allowing the robot's direction to be amended and making the robot go closer to the goal while reducing computation runtime and memory usage. (Xiang Rong and Hong Mei, 2020)

(Cao *et al.*, 2016) use a Genetic Algorithm to obtain the optimal path of an AUV and improve the speed of convergence by enhancing the path planning adaptation function, initial population generation method and the

chamfer operator. The path is demonstrated in the Cartesian coordinates form to display the relative positions of the grid used to find path lengths, but when creating the initial population, a short length encoding of grid serial numbers is used instead for simplicity. The convergence of the Genetic Algorithm is fast if the initial population consists of good routes. (Cao *et al.*, 2016)

(Cao *et al.*, 2016) use randomly generated halfway points to produce paths resulting in non-feasible generated paths. (Cao *et al.*, 2016). overcome this issue by implementing a function that detects if the path between two adjacent points goes through the object, where the algorithm will keep generating random paths until the path does not go through the object. The algorithm obtains the safe path to the target using the adaptive function. (Cao *et al.*, 2016) use two mutation functions one is the standard bit mutation, and the other is a section of a path that is replaced either by a straight line or a randomly generated path.

Chen et al. developed an algorithm combining the Genetic Algorithm and ACO algorithm called GIACO to counter the algorithms' disadvantages. The ACO algorithm has a slow convergence speed due to insufficient starting knowledge of the environment as algorithm initializes the pheromones using a constant, resulting in the ants carrying out blind search. The Genetic Algorithm has an issue of inadequate utilization of heuristic information and observed during simulation, the planning time increased due to iteration redundancy at the later stage of the search. (Chen and Qi, 2020). The key idea is to initialize the Ant Colony pheromone matrix at the beginning of the fusion by using the best path found by the Genetic Algorithm allowing the ants to start with a near-optimal path instead of blindly searching. GIACO applies the advantages of both algorithms to counter their disadvantages. The method applies the Genetic Algorithm's advantage of Global search to counter the blind search problem faced by the standard ACO algorithm and uses the ACO algorithm's advantage of efficient and effective search during the later stage of the search algorithm countering the iteration redundancy faced by Genetic Algorithms. The gradient descent algorithm smooths the generated path. (Chen and Qi, 2020)

GIACO is less diverse when compared to the ACO algorithm, though this may be a benefit as a higher diversity equates to higher randomness, which may decrease search quality. GIACO tackles threat avoidance by dynamically adjusting the step size, which reduces the step size and results in a finer search allowing UAVs to avoid threats more effectively. GIACO reduces computation time, making the algorithm faster than both the Genetic Algorithm and ACO algorithm, and can find an optimal solution in 5 iterations, the path being shorter than its separate counterparts, where it took 12 iterations for ACO and 13 for GA.(Chen and Qi, 2020)

GIACO deals with threats using a dynamic step size which is changed when a sudden threat is detected and reduced when in a volatile area to make the search finer and operate smoothly on the mission, making the GIACO suitable for operation on the battlefield. (Chen and Qi, 2020)

UAVs can fly safely, with better precision than humans, and achieve autonomy using autonomous control systems. There are types of automation ranging from non-autonomy to full autonomy, where a non-autonomous UAV would require a human operator, whereas a UAV with full autonomy does not. Semi-autonomous models would aid the human operator with functionalities such as autopilot and stabilization. (de Moura Souza and Toledo, 2020)

(de Moura Souza and Toledo, 2020) proposed a fusion algorithm combining the Genetic Algorithm with the Ray Casting algorithm to tackle the path planning problem with a safe emergency landing to help UAVs plan paths with full autonomy. The key idea is to use a Genetic Algorithm to produce the best path, which is the chromosome with the highest fitness function value, and the Ray Casting algorithm will detect no-fly zones and landing zones to avoid collisions, implemented to be calculated during the fitness calculation of individuals. (Souza et al. 2020) initialized the Genetic Algorithm using an object that contains an origin, no-fly zones, and a destination, with the output being the best route and modelled using the Cartesian system. Convergence occurs when no new individuals are being inserted into the population, as this occurs only if the child individual has a larger fitness value than one of the parents and the Genetic Algorithm stops when the execution time concludes. (de Moura Souza and Toledo, 2020)

The Ray Casting algorithm, when applied to the path planning problem, prevents collisions intercepting the polygon created by the area of casted horizontal rays from all waypoints on the route, where an even number of interceptions indicate the waypoint is in the area and an odd indicating out of the area. (Souza et al. 2020) use two types of the Ray Casting algorithm where one determines if there is a point within the generated polygon, and the other allows the UAV to determine if any obstacles collide with it and if there are any collisions between the waypoints themselves. The algorithm was deployed on a Raspberry Pi, indicating its lightweight capabilities allowing the algorithm to be suitable for UAVs.(de Moura Souza and Toledo, 2020)

Wang et al. (2021) use a Genetic Algorithm to tackle the path planning and trajectory problem for multi-UAV search. The method also uses chromosomes to model potential continuous paths created after the population is initialized and ranked based on their fitness function score. The size path formula is the Euclidian distance between the target and start positions and is used to retrieve the length of the chromosomes. The objective function, in this case, is the fitness function which is $1/size\ path$, meaning that the smaller the path is, the larger the fitness score. The method can optimize five paths after 50 iterations. The algorithm then initializes the Genetic algorithm using point coding to generate the first generation and then updates each iteration using selection to remove suboptimal paths.(Wang, Jiang and Bao, 2021).

III. METHODOLOGY

This paper creates a fusion algorithm combining the Genetic Algorithm with Ant Colony Optimization based on the idea proposed by (Chen and Qi, 2020) and is applied to path planning and trajectory for dynamic environments to aid UAV data collection.

A. Path Planning Algorithm

Algorithm 1 Fusion Path Planner

```
Initialize population
while termination criteria not reached do
    calculate fitness
    perform selection
    perform crossover
    perform mutation
    evaluate individuals
end while
return best GA path
initialize ACO parameters
while termination criteria not reached do
    set initial ant on GA path
    for each ant after initial do
      assign all nodes as not visited
      while unvisited nodes > 0 do
        use transitional probability to choose next node
        update pheromone matrix with new level
      end while
    if current path < previous path then
      update stored best path with current path
end while
return shortest path
```

Both the GA and ACO algorithms are biologically inspired, where the GA is inspired by the theory of evolution, and the ACO is inspired by Stigmergy, which is the indirect communication of organisms using the environment.(Dorigo, Birattari and Stutzle, 2006; Lamini, Benhlima and Elbekri, 2018). GAs performs well as a global search algorithm that can be applied to various domains; in this case, the Genetic Algorithm is applied to graph-based path planning for UAV data collection. The global search capabilities are used to find a near efficient reference path, which will allow the algorithm to escape from the local optimal path. The second stage of the algorithm uses the Ant Colony Optimization algorithm to refine and optimize the path due to its excellent local search capabilities and efficiency in the TSP.(Lamini, Benhlima and Elbekri, 2018). Ant Colonies use Stigmergy and autocatalysis-based positive feedback to find the shortest path to a food source from the nest, depositing pheromones along the route to create pheromone trails. Ants base their decisions on probability, where the higher the pheromone concentration on a vertex, the more likely it would be for the ant to choose that vertex.(Dorigo, Birattari and Stutzle, 2006).

Like GA, ACO can be applied to a graph network to retrieve the shortest path. Artificial ants will traverse the vertex between nodes while depositing pheromones, where the number of pheromones deposited at an instance depends on the quality of the vertex. Vaporization of pheromones allows the algorithm to remove lower quality paths as higher quality paths would frequently have pheromones deposited at each vertex in the routes resulting in bad paths slowly diminishing, directing ants to follow better paths. (Dorigo, Birattari and Stutzle, 2006). The Ant Colony Optimization algorithm performs well in the Travelling Salesman Problem due to its parallel nature, though it can fall into local optimum (Oonsrikaw and Thammano, 2018).

Both the ACO and Genetic Algorithm perform well in finding optimal paths to the Travelling Salesman Problem, though they both have disadvantages. The Genetic Algorithm suffers from iteration redundancy, and the ACO algorithm suffers from slower convergence due to the initial blind search. (Chen and Qi, 2020). This is solved when combining the two algorithms as the ants could follow the GA path initially instead of randomly traversing the graph, and since the Genetic Algorithm path is not final, fewer iterations are needed as ACO can optimize the path further, which speeds up the runtime as the Genetic Algorithm is slower than Ant Colony Optimization. Runtime is very important, especially in dynamic environments, as the faster, the algorithm can retrieve a path, the more it can be executed in a given time, allowing the UAV to traverse with more relevant information.

The implementation in this paper slightly differs from (Chen and Qi, 2020) as the fusion is of a GA and standard ACO instead of the IACO and also does not initialize the pheromone matrix directly using the path generated from GA. Instead, the pheromone matrix is initialized uniformly, and the initial Ant is set to the Genetic Algorithm path, depositing pheromones on the return, and directing other ants to follow the route as the pheromone count is higher. Though the implementation is slightly different, the fundamental idea is the same combining the GA and ACO to increase performance.

1) Stage 1: Genetic Algorithm

The chromosomes in the genetic algorithm are encoded using strings and are initialized by randomly shuffling the nodes in the environment; this means the solutions will contain feasible routes as there will not be any node duplications. Each generated chromosome will undergo a fitness calculation which is $1/\sqrt{\sum_1^n (yi - xi)^2}$. The Euclidean distance is used to calculate the distances between nodes, which will allow the chromosomes in the population to be ranked based on their score. A higher score will indicate a fitter chromosome, meaning the chromosome will be more optimal with regard to the objective function.

Tournament Selection is implemented due to its linear time complexity and efficiency and always uses chromones with the highest fitness for crossover. The Fusion algorithm also implements elitism to pass the fittest chromosome into the next population to ensure the fittest chromosome is not lost when the crossover is applied. (Fang and Li, 2010).

For reproduction, the One Point Crossover method has been implemented using the middle node in the chromosome as a pivot point. For one of the parent chromosomes, any nodes

after the pivot point are removed and filled from nodes in the second parent chromosome and checked for duplicate nodes.

The mutation function has been implemented by retrieving the nodes in the route and swapping the position of nodes if a randomly generated number is less than the mutation rate. The mutation rate is generally low as a higher mutation rate will result in increased randomness, negatively affecting the solutions.

2) Stage 2: Ant Colony Optimization and GA link

The first step is initializing the pheromone matrix, which is initialized uniformly with a number above zero and below 1. Ants use probability to choose the next node, with a bias towards vertexes with a higher number of deposited pheromones and better-quality vertexes. (Dorigo, Birattari and Stutzle, 2006)

$$p_{ij} = \frac{(\tau_{ij})^{\alpha}(\eta_{ij})^{\beta}}{\sum((\tau_{ij})^{\alpha}(\eta_{ij})^{\beta})} \quad (1)$$

Equation 1 demonstrates how ants decide the next node for traversal. The Tau parameter indicates the pheromone level of the vertex, eta indicates the quality of the vertex, and the parameters alpha and beta dictate the importance of these factors. The numerator is for the specific vertex, whereas the denominator is the sum of the current vertex and all previous vertex in the path.(Dorigo, Birattari and Stutzle, 2006).
The quality of the vertex is demonstrated using the equation:

$$\eta_{ij} = \frac{1}{d_{ij}} \quad (2)$$

Similar to the fitness function for the Genetic Algorithm, 1 over the vertex length as seen in equation 2 will give a higher score when the length is shorter, resulting in a higher probability. This equation is also used to dictate the number of pheromones to be deposited.(Blum, 2005).
The pheromone matrix must be initialized with a number above zero and preferably below 1, as it will lead to a zero division when the transitional probability is called for the first time.(Blum, 2005).
The first ant is initialized using the path generated by the Genetic Algorithm, starting at the first node in the path and updating the matrix on return which increases the probability ants take the route provided by the GA as the pheromones will be higher.
Once the ants have traversed a vertex, they will deposit pheromones and update the pheromone matrix. The number of pheromones deposited is modelled using equation 3:

$$\tau_{ij} = (1-\rho) \times \tau_{ij} + \frac{Q}{l_{ij}} \quad (3)$$

Where the number of pheromones deposited depends on the length of the vertex between nodes where the longer the path, the less pheromones deposited as Q is a constant and li is the vertex length. (Blum, 2005).

Pheromone Vaporization is implemented using the term $(1-\rho)$, where Rho is the evaporation rate. Pheromone Vaporization benefits the Ant Colony Algorithm by slowly diminishing all the pheromone trails making all the longer routes vanish in time as shorter paths are more frequently visited by ants stopping premature convergence to a suboptimal solution as the ants are made to forget earlier suboptimal choices and are encouraged to explore the environment. All the good choices are retained as the more frequently visited shorter paths will always have more pheromones deposited in the vertex. (Dorigo and Stützle, 2004).

The ants will continue to traverse the environment until all specified ants have returned home and the total distance of the current path is compared to the distance of the previously stored path, where it would be replaced only if the distance is shorter and passed to the UAV for trajectory planning once the termination criteria is met.

B. Path Planner applied to UAV for data collection

The environment consists of a 2D grid of 1km by 1km. Nodes are placed onto the grid randomly and move at a constant speed using the levy motion model. The UAV traverses the environment using a defined trajectory that covers all four grid corners. The UAV can move in an 8-way direction when traversing the environment. Nodes can store information about other nodes it has encountered. When the UAV encounters a node, it stores the identification number and coordinates at the time and the information of other node locations. In this paper, the path planner is used to create a detour function that can make the UAV leave the defined trajectory and move toward other nodes. The UAV uses the path planner to return the shortest route between the captured nodes, where the UAV will travel to each node until the path planner is executed again or completes the tour, where the UAV will return to the initial trajectory. If the UAV is on a detour and the energy level is lower than the distance from the base, it will discard its trajectory and return to the base.

IV. Results

All the results presented in the table are each an average of 10 runs, simulated using an Intel® Core™ i7-9700k CPU with 32GB RAM and AMD Ryzen 9 5900HS with 16GM RAM.

A. TSP Performance

TABLE I. SHORTEST PATH PRODUCED ON BERLIN52 DATASET

path planner performance		
Algorithms	shortest path	standard deviation
GA	15831.7	872.867
ACO	9892.630971	327.918
Fusion Algorithm	9026.384	130.639

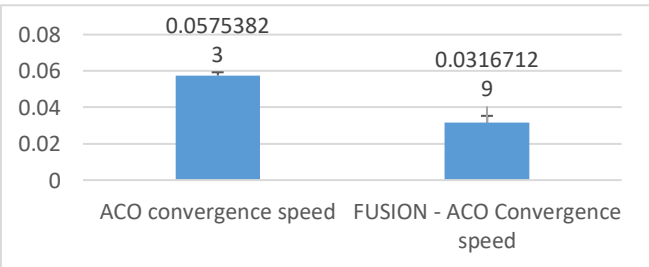

Fig. 1. Convergence speed of ACO for base ACO and fusion

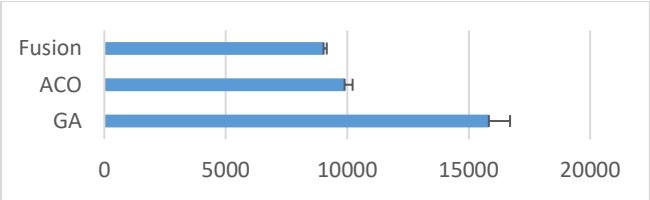

Fig. 2. TSP performance shortest path on Berlin52

The Fusion Algorithm is tested against GA and ACO where Table 1 shows the performance of the three algorithms.

B. *UAV Simulation on dynamic environments*

TABLE II. DIRECT NODE CAPTURES

Direct Node Captures				
	Nodes Captured	Drone Range		
		50m	72m	100m
Initial Trajectory	100 Nodes	32.3	48.5	59.5
	200 Nodes	70.6	90	113.7
Detour	100 Nodes	33.5	48.9	64.2
	200 Nodes	64.3	96.3	118.7

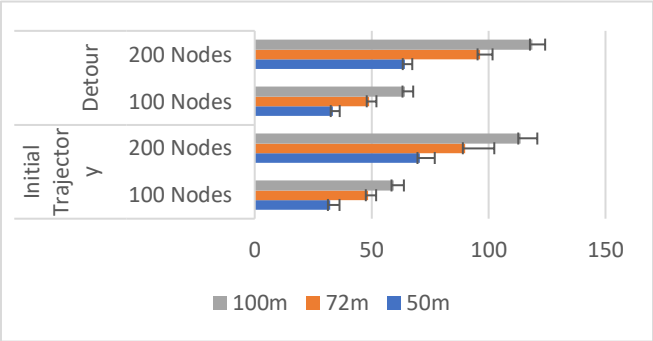

Fig. 3. Direct Node Captures

TABLE III. NODE CAPTURES WITH CACHING

Node Captures with Caching				
	Nodes Captured	Drone Range		
		50m	72m	100m
Initial Trajectory	100 Nodes	42.10	54.50	63.20
	200 Nodes	102.30	111.20	129.90
Detour	100 Nodes	43.00	56.40	72.40
	200 Nodes	94.10	117.70	133.30

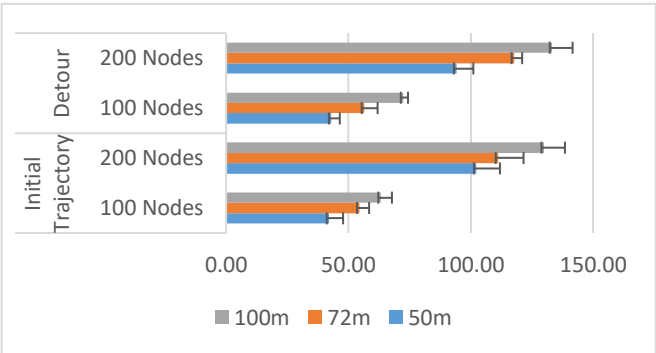

Fig. 4. Node Captures with Caching

The UAV path planning and trajectory simulations have been tested on a 2D array of 1km-by-1km range. The UAV is able to traverse the map through an 8 directional movement function. Though in a real-world scenario the UAV would operate in a 3D plane, the simulation has been conducted on a 2D plane for simplicity and to test the path planning capabilities.

Detouring activates when the UAV captures a certain number of nodes and traverses half of the initial trajectory. The length of the simulations was calculated to make the UAV able to complete its journey using the initial trajectory. When the UAV detours, it will return to base before the end of the simulation. Constraining the simulation time to the time taken to complete a full loop of the initial trajectory simulates the real-world scenario of energy constraints. Since UAVs are limited to the amount of fuel, simulations are run with the assumption that the UAV has maximum energy capacity, which means that if the UAV detects that it is running low on energy while detouring, it will discard its planned path and return to home.

For 100 nodes the detour will become activated after the UAV collects 35 nodes for both ranges 50 and 72 and 60 nodes when range is 100. For 200 nodes the detouring functionality becomes activated when the UAV collects 60 nodes for ranges 50 and 72 and 100 nodes when using a range of 100. These conditions have been implemented to stop the UAV from detouring too early into the simulation as this can lead to the UAV making poor decisions due having less information when detouring.

Direct node captures as seen in Table 2 demonstrates the number of nodes captured when the UAV interacted with a node directly. Nodes are also able to capture and store information of other nodes that it comes into contact with. Node captures with caching as seen in Table 3 demonstrates all the nodes the UAV has come into direct contact with as well as nodes captures by other nodes.

V. DISCUSSION

Table 1 displays the effectiveness of the Fusion algorithm against the base methods GA and ACO in solving TSP using the 52 Berlin dataset. The known optimal solution to the 52 Berlin dataset is 7544.3659 (Abdoun, Tajani and Abouchabka, 2012). The ACO parameters used for both the fusion algorithm and ACO were based on (Mustafa, Altiok and Koçer, 2015) which produced an average path length of 9892.63 with a standard deviation of 327.918 for ACO and 9026.34 with a standard deviation of 130.639 for the Fusion algorithm.

To compare the performance of the Fusion Algorithm against GA, ACO, The GA parameters are used for the GA aspect of the Fusion algorithm and the ACO parameters are used for the ACO aspect of the Fusion algorithm. ACO produces a path 37.51% shorter than GA suggesting that ACO is more effective at TSP than GA. The fusion algorithm produces a path 42.98% shorter than GA suggesting that combining ACO to GA has a positive impact on the effectiveness of the algorithm. The fusion algorithm produces a path 8.76% shorter than ACO. This suggests that the GA has a positive impact on the performance of ACO when combined benefiting from a globally searched solution. The GIACO

implementation by (Chen and Qi, 2020) which this method takes inspiration from also outperforms both GA and ACO which further reinforces the superior performance of combining GA with ACO.

Figure 1 shows the speed of the two algorithms when applied to TSP. (Chen and Qi, 2020) stated the GA improves the convergence speed of ACO due to mitigating the blind search problem, this can also be seen as the ACO stage of the Fusion algorithm completes 44.9% faster than the standard ACO algorithm, which further suggests GA improves the performance of ACO.

Tables 2 and 3 demonstrate the performance of the UAV using the fusion algorithm as a path planner for detouring against a base square trajectory that covers all four corners of the environment. Both Tables 2 and 3 show a positive correlation on the number of nodes captured as the communication range is increased, which the most noticeable improvement for direct node captures happens when moving from a communication range of 50m to 72m for both initial and detour enabled trajectories with 50.1% more captures nodes using the initial trajectory and 45.97% increase in node captures when using the detour enabled trajectory using 100 nodes. The relation is also the same when using 200 nodes though, this time the detour enabled trajectory benefits more from the increased range than the initial trajectory with a node capture increase of 49.8% as opposed to the initial trajectory of 27.47%. This may be due to the path planner returning a path with more nodes allowing the UAV to make more detours than when testing against 100 nodes. UAV captures with caching also show a similar trend when testing with 100 nodes, although when testing with 200 nodes moving from a communication range of 72m to 100m showed the most improvement with 16.81% increase in capture for the UAV using the initial trajectory and 13.25% increase when using detour enabled trajectory. Increasing the communication range affects the detour enabled trajectory more than the initial trajectory for direct captures at 100 nodes but affects initial trajectory more for caching enabled captures with 200 nodes as doubling the communication range increases the direct node captures by 91.6% when detour enabled as opposed to 84.21% when not in use for 100 nodes similarly when testing with 200 nodes the increase in node capture is 81.58% when detour is enabled and 61 % when not in use. The same trend is seen for UAV node captures with caching enabled.

Doubling the number of nodes from 100 to 200 does not double the number of nodes captured, it can be seen from Table 2 that the highest direct node capture increase when moving from 100 to 200 nodes was achieved by the UAV using the initial trajectory with a communication range of 50m of 18.57% more nodes.

Detour enabled trajectory has shown to be more effective than the initial trajectory when increasing the communication range as the detour enabled trajectory is able to directly capture 7.89% more nodes when using a communication range of 100m than the initial trajectory and also 14.56% increase in node captures with caching enabled for 100 nodes. For the ranges 50 and 72 the performance is similar between both methods which is also projected with node caching enabled. When testing with 200 nodes the detour enabled trajectory performs worse than the initial trajectory when using a communication range of 50m, where the initial trajectory directly captures 8.92% more nodes than the detour enabled trajectory and 8.01% more nodes with caching enabled. When the communication range is increased the detour enabled trajectory outperforms the initial trajectory, but by at most 7% direct node captures and 5.8% node captures with caching for 200 nodes. This suggests that the detour enabled trajectory is effective when the communication range is higher, which may be due to the UAV having more information to make better decisions on the location to detour.

VI. CONCLUSION/ FUTURE WORK

In this paper, a path planning algorithm was created from the fusion of the Genetic Algorithm and Ant Colony Optimization algorithm based on (Chen and Qi, 2020) using the Ant Colony Algorithm to mitigate iteration redundancy and using the Genetic Algorithm to mitigate blind searching which occurs at the start of the Ant Colony Optimization (Chen and Qi, 2020). The Fusion algorithm benefits from global search capabilities and Ant Colony Optimization efficiency, demonstrated in Table 1, outperforming the GA and ACO in producing the shortest path. This paper takes path planning further by testing the performance of the fusion algorithm in a dynamic environment where nodes are also traversing the field and has shown to improve upon the initial trajectory when the UAV communication range is higher, as seen in Tables 2 and 3 though can be improved to also perform better with low communication ranges.

Detouring does not perform well when the path planner is activated early in the simulation. Further work is required to develop a better decision and situational awareness function that allows the UAV to be more autonomous and efficient. In this paper, the proposed method moves towards nodes, which means it can end up following nodes, which can be improved by implementing node interception by considering the distance from the UAV and the direction the node travels in.

ACKNOWLEDGMENT

I would like to thank Umair Bilal Chaudhry for his support and advice during this project.

REFERENCES

Abdoun, O., Tajani, C. and Abouchabka, J. (2012) “Hybridizing PSM and RSM Operator for Solving NP-Complete Problems: Application to Travelling Salesman Problem,” *arXiv:1203.5028 [cs]* [Preprint]. Available at: https://arxiv.org/abs/1203.5028.

Blum, C. (2005) “Ant colony optimization: Introduction and recent trends,” *Physics of Life Reviews*, 2, pp. 353–373. Available at: https://doi.org/10.1016/j.plrev.2005.10.001.

Cao, J. *et al.* (2016) “Genetic-Algorithm-Based Global Path Planning for AUV,” pp. 79–82. Available at: https://doi.org/10.1109/ISCID.2016.2027.

Chen, X. and Qi, L. (2020) “UAV Path Planning Based on The Fusion Algorithm of Genetic and Improved Ant Colony,” pp. 307–312. Available at: https://doi.org/10.1109/CAC51589.2020.9326555.

Dorigo, M., Birattari, M. and Stutzle, T. (2006) “Ant colony optimization,” *IEEE Computational Intelligence Magazine*, 1, pp. 28–39. Available at: https://doi.org/10.1109/MCI.2006.329691.

Dorigo, M. and Stützle, T. (2004) *Ant Colony Optimization*. Mit Press.

Fang, Y. and Li, J. (2010) “A Review of Tournament Selection in Genetic Programming,” *Advances in Computation and Intelligence*, 6382, pp. 181–192. Available at: https://doi.org/10.1007/978-3-642-16493-4_19.

Gong, J. *et al.* (2018) “Flight Time Minimization of UAV for Data Collection Over Wireless Sensor Networks,” *IEEE Journal on Selected Areas in Communications*, 36, pp. 1942–1954. Available at: https://doi.org/10.1109/JSAC.2018.2864420.

Goudarzi, S. *et al.* (2019) “Data collection using unmanned aerial vehicles for Internet of Things platforms,” *Computers & Electrical Engineering*, 75, pp. 1–15. Available at: https://doi.org/10.1016/j.compeleceng.2019.01.028.

Lamini, C., Benhlima, S. and Elbekri, A. (2018) “Genetic Algorithm Based Approach for Autonomous Mobile Robot Path Planning,” *Procedia Computer Science*, 127, pp. 180–189. Available at: https://doi.org/10.1016/j.procs.2018.01.113.

de Moura Souza, G. and Toledo, C.F.M. (2020) “Genetic Algorithm Applied in UAV’s Path Planning,” pp. 1–8. Available at: https://doi.org/10.1109/CEC48606.2020.9185909.

Mustafa, Altiok and Koçer, B. (2015) “The Analysis of GR202 and Berlin 52 Datasets by Ant Colony Algorithm,” pp. 103–108. Available at: https://doi.org/10.1109/ACSAT.2015.47.

Nagib, G. and Gharieb, W. (2004) “Path planning for a mobile robot using genetic algorithms,” *International Conference on Electrical, Electronic and Computer Engineering, 2004. ICEEC ’04.*, pp. 185–189. Available at: https://doi.org/10.1109/iceec.2004.1374415.

Oonsrikaw, Y. and Thammano, A. (2018) “Enhanced Ant Colony Optimization with Local Search,” pp. 291–296. Available at: https://doi.org/10.1109/ICIS.2018.8466388.

Wang, S., Jiang, Z. and Bao, X. (2021) “Autonomous Trajectory Planning Method for Multi-UAV Collaborative Search,” pp. 84–88. Available at: https://doi.org/10.1109/ICACR53472.2021.9605170.

Xiang Rong, T. and Hong Mei, X. (2020) “Robot path planning based on A algorithm and genetic algorithm,” pp. 101–105. Available at: https://doi.org/10.1109/ICCEIC51584.2020.00028.

Xu, Y. and Che, C. (2019) “A Brief Review of the Intelligent Algorithm for Traveling Salesman Problem in UAV Route Planning,” pp. 1–7. Available at: https://doi.org/10.1109/ICEIEC.2019.8784651.

Yin, Youlong *et al.* (2021) “Multi UAV cooperative task allocation method for intensive corridors of transmission lines inspection,” pp. 203–208. Available at: https://doi.org/10.1109/CEI52496.2021.9574502.

MSc Project - Reflective Essay

Project Title:	Path Planning and Trajectory for UAVs
Student Name:	**Tarek Ahmed**
Student Number:	**170431981**
Supervisor Name:	Umair Bilal Chaudhry
Programme of Study:	MSc Artificial Intelligence

This project demonstrates the performance of using path planning to improve the ability of UAVs to collect data from mobile moving stations when compared against an initial trajectory. The project is based solely on software, trying to simulate real world scenarios as close as possible though the simulation is based on a 2D graph approach as opposed to the 3D environments UAVs would travel in for the real world. This decision has been taken to reduce complexity and execution time and to test the performance of the path planner against the initial trajectory. In this reflective essay the strength and weakness are discussed as well as future work, sustainability, and comparison of theoretical performance on practical performance.

- Analysis of strengths/weaknesses

Initially a Genetic Algorithm was going to be used for path planning to aid the UAV to take detours, while looking at literature it was found that the Genetic Algorithm can be slow and lead to iteration redundancy and it was found that the Ant Colony Algorithm is effective at TSP though can fall into local optimum (Chen and Qi, 2020). In this project the idea from (Chen and Qi, 2020) was used to recreate an algorithm that also combines the Genetic Algorithm with ACO though instead of initialising the pheromone matrix as done by (Chen and Qi, 2020) in this implementation the first ant traverses the environment using the path generated from the Genetic Algorithm updating the pheromone matrix on return and uses the standard ACO instead of the IACO used in the (Chen and Qi, 2020) paper, both have shown improvements to the base algorithms.

The path planner was successfully integrated into the UAV simulator where it returns a path when the UAV initiates a pathfinding search. The UAV can now traverse in 8 directions instead of 4 and can take detours based on information provided from the path planner and the creation of a simple decision function that takes into consideration the number of nodes and distance travelled by the UAV before making detours, this allows the UAV to not detour too early as that can lead to the UAV becoming stuck in a given area. The UAV also discards any nodes it has visited and does not visit it again until the next path plan scan, to make sure the UAV does not make unnecessary detours to nodes it has already seen, though this has been set to reset when the path planner executes a new scan as nodes may have cached more information within that time.

This project has shown that path planning can be used in dynamic environments to improve the efficiency of UAVs in their task to collect data from nodes though there are some weaknesses, the main problem being when the UAV collects low amounts of nodes and detours it can cause performance decrease as it would traverse around fewer nodes where it could gain less information, this was prevented by making the UAV only start detouring when certain number of nodes was collected and traversed half of the initial trajectory, a better method should be implemented to make the UAV more autonomous in decision-making. Another weakness is the path planner returns the shortest route between nodes without taking into consideration the current position of the UAV, this would at times make the UAV travel across the environment to scan a node, although this may help in capturing nodes along the way, it isn't an efficient way to detour. Since the project was purely software based environmental factors such as wind speed, temperature and UAV hardware health was not taken into consideration, as would if tested on hardware.

- Presentation of possibilities for further work and Work that you would have conducted if you had more time.

Due to the constraints of time for this project, ideas such as node interception was not able to be implemented, instead of chasing nodes while they are also moving, it would be better for the UAV to use the path planner to predict where node will be based on its current position allowing the UAV to meet the node faster as they would be going towards each other. An implementation of a GUI to display nodes and drone movement would allow to see how the UAV is interacting with the nodes and the trajectory it has taken; this would give the developer a better understanding of how the UAV is operating. Testing of different scenarios such as how fast the UAV can collect a specified number of nodes using detouring and compare against the initial trajectory. Testing the performance of the UAV for both detouring and initial trajectory when supplied with half the maximum energy to complete a full course. If there was more time, a better decision function would be worked on to allow the UAV to be more adaptive when deciding to detour using time and environmental factors. Also adding obstacles to the environment to experiment

the effects of UAV decision - making when detouring. Another idea that is possible for future work would be to take into consideration nodes that take longer to transmit data where the UAV would choose to visit depending on the charge available or could instead visit other nodes that transmit faster.

- Critical analysis of the relationship between theory and practical work produced

This project proposes a novel approach which combines the testing of path planners and UAV data collection capabilities. Both the path planner and simulator are created in java where the inspiration of the path planner was based on (Chen and Qi, 2020) using the path generated from the Genetic Algorithm to aid the search of the Ant Colony Algorithm. (Chen and Qi, 2020) demonstrated that GIACO had superior performance on static path planning when compared against ACO and GA, this is also the case for this implementation when tested against the proposed path planning algorithm, as seen in table 1 where the proposed algorithm is both faster than GA and ACO and in figure 1 the experiments shows that the ACO for the fusion algorithm is 44% faster than the standard ACO algorithm, which goes in line with (Chen and Qi, 2020) as they also stated that by combining the GA to the ACO it will speed up the ACO as it wouldn't be blindly searching initially.

In this project path planner has been taken further and applied to UAV data collection where it is used to indicate to the UAV the trajectory of the detour. The path planner performs TSP on the nodes captured by the drone allowing the UAV to detour and visit the nodes to scan for any surrounding nodes in the area. Initially it was thought that detouring would improve the UAVs ability to capture nodes, but it was seen that when the UAV detours too early in the simulation it performs worse than the initial trajectory, this may be because early in the simulation nodes may have not come across other nodes resulting in them containing less information than they would as the simulation continues, resulting in the UAV traversing to these nodes and not gaining any new information from them and also wasting time where the defined trajectory would just follow the predefined route capturing any new nodes in its path. This was fixed by making the UAV detour after traversing half of the defined trajectory journey and also if it caches a certain number of nodes, this had shown to improve the performance of the path planner enabled detour trajectory and for

100 nodes outperformed the initial trajectory by directly capturing 7.89%% more nodes when the communication range is 100m, but the performance gain is less when the communication range is smaller with 3.7% more nodes with a communication range of 50m and no improvement in direct node capturing when range is 72m. When tested with 200 nodes the initial trajectory performs better when the communication range is smaller as seen in tables 2 and 3 directly capturing 8.9% more nodes than the detour trajectory but as the range is increased the detour trajectory outperforms the initial trajectory capturing 7% more nodes when the communication range is 72m and 4.39% more nodes when the communication range is 100m. The detour trajectory performed relatively better when tested against 100 nodes where the initial trajectory actually beat the detour trajectory when the communication range was 50m with 200 nodes on the environment.

This may suggest that the implementation of the detour constrains are more effective with less nodes and would need to be worked on to also show an improvement on more nodes in the environment as the relative performance with 100 nodes is greater. Each value on the tables is an average of 10 runs, based on this, it can be concluded that when the communication range is higher, the path planner enabled detour trajectory is more effective with effectiveness decreasing as the range decreases.

- Awareness of Legal, Social Ethical Issues and Sustainability

The proposed method is eco-friendly for sustainability issues as it has shown to capture more nodes in a given run cycle which when demonstrated as a real-world problem, by making the UAV capture more information using the same amount of energy, it allows for the UAV to have to use less energy to capture certain number of nodes, this would help reduce total energy consumption in a given time frame. The path planning algorithm has been used to test performance on UAVs for data collection though ethical issues may arise if the method is used for surveillance such as conflicts of interest, intrusion, and discrimination.(Kan, 2020)

- Contribution
  - For this project a new path planning algorithm was created which was inspired by (Chen and Qi, 2020)
  - Integrated the path planner into the simulator
  - Created the 8-way movement system
  - Created a method to allow the UAV to start detouring once it traversed on the initial path for some distance and collected a certain number of nodes.
  -
- Personal development

Prior to the project I had limited knowledge of genetic algorithms, path planning in general and also UAV aided data collection. This project has given me a good understanding of path planning and how it can be used to make UAVs more autonomous. The core functionality has been successfully implemented which allows the UAV to plan for the shortest path and take detours depending on the amount of energy it has and the number of nodes it has also captured. Prior to taking an MSc in Artificial Intelligence my coding experience was fairly low having started programming only after starting my bachelor's degree, though in this time I have learned more about data structures, algorithms and have been able to improve my problem-solving skills while undertaking this project and degree.

- Conclusion

In this essay the strengths and weaknesses of the project and future work has been discussed, the contributions have been stated. It has also been shows that when using a higher range, the proposed method is superior to the predefined trajectory, though there are real world factors such as wind speed, heat and hardware health that has not been taken into consideration, which can be tested with hardware in real time for real world practicality.

References

Chen, X. and Qi, L. (2020) "UAV Path Planning Based on The Fusion Algorithm of Genetic and Improved Ant Colony," pp. 307–312. Available at: https://doi.org/10.1109/CAC51589.2020.9326555.

Kan, T.L.J. (2020) "Ethical issues in the use of surveillance cameras to support ageing in place," *summit.sfu.ca* [Preprint]. Available at: https://summit.sfu.ca/item/21085#:~:text=Nine%20ethical%20issues%20emerged%20from.